

5) Suggested KPIs-Features-Metrics-Statistics that could be added in the SSAS

COMPREHENSIVE SSAS CUBE DESIGN: MEASURES, DIMENSIONS & KPIs

Complete List of Features, Metrics, and Aggregations for Your Data Cube

INTRODUCTION

This document provides a comprehensive list of all **measures, dimensions, KPIs, and calculated metrics** that can be extracted from your dataset and implemented in SSAS (SQL Server Analysis Services). These will form the foundation of your data cube for Power BI reporting.

PART 1: DIMENSIONS (Attributes for Filtering & Grouping)

Dimensions are the **"by what"** categories—the attributes you use to slice and dice your data.

1.1 Student Dimensions

Dimension	Source Column	Type	Description	Values
Student Level	StudentLevel	Attribute	Academic level of student	Undergraduate, Graduate, PhD
Student Discipline	Discipline	Attribute	Subject area of study	7 values
Student Segment	Derived	Attribute	Combination of Level + Discipline	$3 \times 7 = 21$ combinations

1.2 Session Dimensions

Dimension	Source Column	Type	Description	Values
Session ID	SessionID	Attribute	Unique session identifier	10,000+ values
Session Date	SessionDate	Hierarchy	Date of session	Year → Quarter → Month → Day
Session Time	Derived	Attribute	Time of day	Morning, Afternoon, Evening, Night
Day of Week	Derived	Attribute	Day of week	Monday, Tuesday, ..., Sunday
Weekday/Weekend	Derived	Attribute	Is it weekend?	Weekday, Weekend
Academic Period	Derived	Attribute	Semester/term	Fall 2024, Spring 2025, etc.

1.3 Activity Dimensions

Dimension	Source Column	Type	Description	Values
Task Type	TaskType	Attribute	Type of learning activity	6 values
AI Level	AI_AssistanceLevel	Attribute	Level of AI assistance	Level 0-4 (ordered)
AI Tier	Derived	Attribute	Grouped AI levels	Low (0-1), Medium (2-3), High (4)

Dimension	Source Column	Type	Description	Values
Prompt Category	Derived	Attribute	Grouped prompt counts	Low, Medium, High
Outcome Status	FinalOutcome	Attribute	Session outcome	4 values
Return Status	UsedAgain	Attribute	Whether student returned	Yes/No

1.4 Session Length Dimensions

Dimension	Source Column	Type	Description	Values
Session Length Bin	Derived	Attribute	Grouped session length	0-15min, 15-30min, 30-45min, 45-60min, 60+min
Satisfaction Tier	Derived	Attribute	Grouped satisfaction	Low (1-2), Medium (3), High (4-5)

1.5 Time Dimensions

Dimension	Source Column	Type	Description	Values
Session Month	Derived	Attribute	Month of session	January-December
Session Quarter	Derived	Attribute	Quarter of year	Q1, Q2, Q3, Q4
Session Year	Derived	Attribute	Year of session	2024, 2025, etc.
Academic Month	Derived	Attribute	Month in academic calendar	1-12 (relative)



PART 2: MEASURES (Numeric Aggregations)

Measures are the **"what"** that you aggregate—the numeric values you sum, average, count, etc.

2.1 Count-Based Measures

Measure	Source	Aggregation	Description	Example C
Total Sessions	SessionID	Count	Total number of sessions	COUNT(Se
Unique Students	Derived	Distinct Count	Number of unique students	COUNT(DI StudentID)
Total Prompts	TotalPrompts	Sum	Total number of prompts generated	SUM(TotalF
Total Session Minutes	SessionLengthMin	Sum	Total minutes across all sessions	SUM(Sessi
Total Tasks Completed	FinalOutcome	Count	Number of completed assignments	COUNT(Fir = 'Comple
Total Task Failures	FinalOutcome	Count	Number of failed assignments	COUNT(Fir = 'Failed')
Total Withdrawals	FinalOutcome	Count	Number of withdrawals	COUNT(Fir = 'Withdraw
Returning Students	UsedAgain	Count	Number who returned	COUNT(Us True)
Non-Returning Students	UsedAgain	Count	Number who did not return	COUNT(Us False)

2.2 Average-Based Measures

Measure	Source	Aggregation	Description	Example Calculation
Avg Session Length	SessionLengthMin	Average	Average session duration	AVG(SessionLengthMin)
Avg Prompts per Session	TotalPrompts	Average	Average prompts per session	AVG(TotalPrompts)
Avg Satisfaction	SatisfactionRating	Average	Average satisfaction rating	AVG(SatisfactionRating)
Avg AI Level	AI_AssistanceLevel	Average	Average AI assistance level	AVG(AI_AssistanceLevel)
Avg Sessions per Student	Derived	Average	Average sessions per student	COUNT(DISTINCT StudentID) / COUNT(DISTINCT SessionID)
Avg Prompts per Student	TotalPrompts	Average	Average prompts per student	SUM(TotalPrompts) / COUNT(DISTINCT StudentID)
Avg Session Length per Student	SessionLengthMin	Average	Average minutes per student	SUM(SessionLengthMin) / COUNT(DISTINCT StudentID)

2.3 Rate & Ratio Measures (KPIs)

Measure	Source	Calculation	Description
Completion Rate	FinalOutcome	COUNT(Completed) / COUNT(Sessions)	Percentage of sessions completed
Success Rate	FinalOutcome	COUNT(Completed OR Passed) / COUNT(Sessions)	Percentage of successful sessions

Measure	Source	Calculation	Desc
Retention Rate	UsedAgain	COUNT(UsedAgain = True) / COUNT(Sessions)	Perce of stu who r
AI Adoption Rate	AI_AssistanceLevel	COUNT(AI_AssistanceLevel > 0) / COUNT(Sessions)	Perce using
High AI Adoption	AI_AssistanceLevel	COUNT(AI_AssistanceLevel >= 3) / COUNT(Sessions)	Perce using
Satisfaction Rate	SatisfactionRating	COUNT(SatisfactionRating >= 4) / COUNT(Sessions)	Perce satisfi
Dissatisfaction Rate	SatisfactionRating	COUNT(SatisfactionRating <= 2) / COUNT(Sessions)	Perce dissai
Prompt Density	Derived	AVG(TotalPrompts / SessionLengthMin)	Prom minut (effici
Sessions per Student	Derived	COUNT(SessionID) / COUNT(DISTINCT StudentID)	Avera sessio per st
Task Completion Rate	FinalOutcome	COUNT(Completed) / COUNT(TaskType)	Perce of tas comp
Return Rate	UsedAgain	COUNT(UsedAgain=True) / COUNT(DISTINCT StudentID)	Stude who reterr
AI Efficiency Rate	Derived	SUM(Completed WHERE AI_Level>0) / SUM(Total AI Sessions)	AI effect

PART 3: KEY PERFORMANCE INDICATORS (KPIs)

3.1 Student Success KPIs

KPI	Calculation	Target	Business Meaning
Overall Success Rate	(Completed + Passed) / Total Sessions	> 75%	Students achieving learning objectives
Student Retention Rate	UsedAgain = True / Total Students	> 40%	Students returning for more
Satisfaction Score	AVG(SatisfactionRating)	> 4.0	Student satisfaction level
Task Completion Rate	Completed / Total Tasks	> 80%	Tasks completed vs. started
Student Engagement Score	AVG(SessionLengthMin)	> 30 min	Time invested by students
Learning Efficiency	Completed / Avg Session Length	Higher = Better	Efficiency of learning

3.2 AI Effectiveness KPIs

KPI	Calculation	Target	Business Meaning
AI Adoption Rate	AI_Level > 0 / Total Sessions	> 60%	AI usage across sessions
AI Success Lift	Success Rate(High AI) - Success Rate(Low AI)	> 15%	AI effectiveness improvement
AI Satisfaction Lift	Satisfaction(High AI) - Satisfaction(Low AI)	> 0.5	AI satisfaction improvement
AI Retention Impact	Retention(With AI) - Retention(Without AI)	> 10%	AI impact on retention
Optimal AI Level	AI_Level with Highest Success Rate	3-4	Best AI assistance level

3.3 Engagement KPIs

KPI	Calculation	Target	Business Meaning
Avg Session Length	AVG(SessionLengthMin)	> 30 min	Typical engagement time
Avg Prompts per Session	AVG(TotalPrompts)	> 5	Interaction depth
Max Session Length	MAX(SessionLengthMin)	-	Peak engagement
Prompt Density	TotalPrompts / SessionLengthMin	0.2-0.5	Activity pace
Session Frequency	Sessions per Student / Month	> 2	Student activity level

3.4 Operational KPIs

KPI	Calculation	Target	Business Meaning
Total Sessions	COUNT(SessionID)	-	Overall platform usage
Active Students	Students with > 3 sessions	-	Engaged student base
Peak Usage Hours	Sessions per Hour	-	When platform is used most
Weekly Active Rate	Weekly Sessions / Total Students	> 20%	Regular engagement



PART 4: CALCULATED MEASURES (Derived)

These measures require calculations within SSAS.

4.1 Moving Averages

Measure	Calculation	Description
7-Day Moving Avg Satisfaction	AVG(SatisfactionRating, Window=7 days)	Weekly trend in satisfaction
30-Day Moving Avg Session Length	AVG(SessionLengthMin, Window=30 days)	Monthly trend in engagement
Rolling Retention Rate	UsedAgain Rate over Last 3 Months	Recent retention trend

4.2 Comparative Measures

Measure	Calculation	Description
YoY Growth - Sessions	(Sessions Current Year - Sessions Previous Year) / Sessions Previous Year	Year over year growth
MoM Change - Retention	(Retention Current Month - Retention Previous Month) / Retention Previous Month	Month over month change
Period Over Period Comparison	Current Period vs Previous Period	Compare any time periods

4.3 Compound Measures

Measure	Calculation	Description
Student Lifetime Value (LTV)	Avg Sessions × Avg Retention × Value per Session	Estimated student value
AI ROI	(Cost of AI) / (Improvement in Retention × Lifetime Value)	Return on AI investment
Student Health Score	(Retention × Satisfaction × Engagement) / 3	Overall student health
Risk Score	1 - Success Rate × Retention Rate	Student risk indicator

PART 5: HIERARCHIES & CALCULATED COLUMNS

5.1 Date Hierarchy

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Level 0: Year (2024, 2025)
└─ Level 1: Quarter (Q1, Q2, Q3, Q4)
    └─ Level 2: Month (January–December)
        └─ Level 3: Day (1–31)
            └─ Level 4: Session Date (YYYY-MM-DD)
```

5.2 Student Hierarchy

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Level 0: Academic Level (Undergraduate, Graduate, PhD)
└─ Level 1: Discipline
    └─ Level 2: Student ID
```

5.3 Activity Hierarchy

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Level 0: Task Type
└─ Level 1: AI Assistance Level
    └─ Level 2: Outcome Status
        └─ Level 3: Session ID
```

5.4 Time-of-Day Hierarchy

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Level 0: Time Period (Morning, Afternoon, Evening, Night)
└─ Level 1: Hour (0–23)
    └─ Level 2: Session Date
```

5.5 Calculated Columns

Column Name	Calculation	Purpose
SessionMonth	MONTH(SessionDate)	Month-based grouping
SessionQuarter	QUARTER(SessionDate)	Quarter-based

Column Name	Calculation	Purpose
		grouping
SessionYear	YEAR(SessionDate)	Year-based grouping
DayOfWeek	DATEPART(weekday, SessionDate)	Weekday analysis
IsWeekend	CASE WHEN DayOfWeek IN (6,7) THEN 1 ELSE 0 END	Weekend filtering
SessionHour	HOUR(SessionDateTime)	Time-of-day analysis
TimeOfDay	CASE WHEN Hour < 12 THEN 'Morning' WHEN Hour < 17 THEN 'Afternoon' WHEN Hour < 21 THEN 'Evening' ELSE 'Night' END	Time period segmentation
LengthCategory	CASE WHEN SessionLengthMin < 15 THEN 'Short' WHEN < 30 THEN 'Medium' WHEN < 45 THEN 'Long' ELSE 'Very Long' END	Session length grouping
PromptCategory	CASE WHEN TotalPrompts < 3 THEN 'Low' WHEN < 6 THEN 'Medium' ELSE 'High' END	Prompt count grouping
SatisfactionTier	CASE WHEN SatisfactionRating < 2.5 THEN 'Low' WHEN < 4.0 THEN 'Medium' ELSE 'High' END	Satisfaction grouping
AITier	CASE WHEN AI_AssistanceLevel < 2 THEN 'Low' WHEN < 4 THEN 'Medium' ELSE 'High' END	AI assistance grouping
PromptDensity	TotalPrompts / SessionLengthMin	Efficiency metric
SuccessFlag	CASE WHEN FinalOutcome = 'Completed' THEN 1 ELSE 0 END	Binary success indicator

PART 6: SCENARIO & TRACKING METRICS

6.1 Retention Tracking

Metric	Calculation	Tracking Period
D1 Retention	UsedAgain next day	Daily
D7 Retention	UsedAgain within 7 days	Weekly
D30 Retention	UsedAgain within 30 days	Monthly
D90 Retention	UsedAgain within 90 days	Quarterly

6.2 Progress Tracking

Metric	Calculation	Tracking Period
Average Tasks Completed per Student	Completed / Unique Students	Per student
Session Progression Rate	Sessions per Week / Student	Weekly
Completion Momentum	Completed Rate Trend	Monthly

6.3 AI Performance Tracking

Metric	Calculation	Tracking Period
AI Usage Trend	AI_Level over time	Monthly
AI Success by Task	Success Rate per Task × AI Level	Task-based
AI Satisfaction Trend	Satisfaction by AI Level over time	Monthly



PART 7: SSAS IMPLEMENTATION SUMMARY

Recommended Dimensions

Dimension	Source	Type	Attributes
Student	StudentLevel, Discipline	Standard	Level, Discipline, Segment
Session	SessionID, SessionDate	Standard	ID, Date, Length, Time
Activity	TaskType, AI_AssistanceLevel	Standard	Task, AI Level, AI Tier
Outcome	FinalOutcome, UsedAgain	Standard	Outcome Status, Return Status
Date	SessionDate	Time	Year, Quarter, Month, Day
Time	Derived	Standard	Hour, Time Period
Satisfaction	SatisfactionRating	Standard	Rating, Tier

Recommended Measures

Type	Measures
Count	Total Sessions, Total Students, Total Prompts
Sum	Total Minutes, Total Completed, Total Returns
Average	Avg Session Length, Avg Satisfaction, Avg Prompts
Rates	Success Rate, Retention Rate, Completion Rate
Percentiles	P90 Session Length, P95 Prompts

Recommended KPIs

Category	KPIs
Success	Overall Success Rate, Task Completion Rate
Retention	Student Retention Rate, Return Rate
Satisfaction	Avg Satisfaction, Satisfaction Rate
Engagement	Avg Session Length, Avg Prompts, Session Frequency
AI	AI Adoption Rate, AI Success Lift, AI Satisfaction Impact

Recommended Perspectives

Perspective	Focus	Key Metrics
Executive	High-level performance	Retention Rate, Success Rate, Satisfaction
Academic	Learning outcomes	Completion Rate, Success Rate by Discipline
AI Performance	AI effectiveness	AI Adoption, AI Lift, Optimal AI Level
Student Engagement	Activity patterns	Avg Session Length, Session Frequency
Operational	Usage and efficiency	Total Sessions, Peak Usage, Active Students



QUICK REFERENCE CARD

Element	Count	Example
Dimensions	7+	Student Level, Discipline, Date
Measures	25+	Avg Session Length, Satisfaction Rate
KPIs	15+	Retention Rate, AI Success Lift
Calculated Columns	12+	AITier, PromptCategory, SuccessFlag
Hierarchies	4+	Date (Year → Quarter → Month)
Perspectives	4+	Executive, Academic, AI Performance

Bottom Line: This comprehensive SSAS design provides complete analytical capabilities across all dimensions of your data, from student success to AI effectiveness to operational metrics. 🎯

Basic Session Metrics:

- Session Count = COUNT(SessionID)
- Total Session Length (sum) = SUM(SessionLengthMin)
- Average Session Length = AVG(SessionLengthMin)
- Min Session Length = MIN(SessionLengthMin)
- Max Session Length = MAX(SessionLengthMin)
- Standard Deviation of Session Length = STDEV(SessionLengthMin)
- Total Prompts (sum) = SUM(TotalPrompts)
- Average Prompts per Session = AVG(TotalPrompts)
- Min Prompts per Session = MIN(TotalPrompts)
- Max Prompts per Session = MAX(TotalPrompts)
- Average Satisfaction Rating = AVG(SatisfactionRating)
- Min Satisfaction Rating = MIN(SatisfactionRating)
- Max Satisfaction Rating = MAX(SatisfactionRating)
- Satisfaction Rating Standard Deviation = STDEV(SatisfactionRating)
- AI Assistance Level Average = AVG(AI_AssistanceLevel)
- AI Assistance Level Mode = MODE(AI_AssistanceLevel)

Time-Based Metrics:

- Sessions per Day = COUNT(SessionID) GROUP BY SessionDate
- Sessions per Week = COUNT(SessionID) GROUP BY DATEPART(WEEK, SessionDate)
- Sessions per Month = COUNT(SessionID) GROUP BY DATEPART(MONTH, SessionDate), DATEPART(YEAR, SessionDate)
- Sessions per Quarter = COUNT(SessionID) GROUP BY DATEPART(QUARTER, SessionDate), DATEPART(YEAR, SessionDate)
- Sessions per Year = COUNT(SessionID) GROUP BY DATEPART(YEAR, SessionDate)

- Average Sessions per Day of Week = $\text{AVG}(\text{COUNT}(\text{SessionID})) \text{ GROUP BY DATENAME}(\text{WEEKDAY}, \text{SessionDate})$
- Peak Day of Week = $\text{TOP 1 DATENAME}(\text{WEEKDAY}, \text{SessionDate}) \text{ ORDER BY COUNT}(\text{SessionID}) \text{ DESC}$
- Peak Month = $\text{TOP 1 DATEPART}(\text{MONTH}, \text{SessionDate}) \text{ ORDER BY COUNT}(\text{SessionID}) \text{ DESC}$
- Average Session Length by Day of Week = $\text{AVG}(\text{SessionLengthMin}) \text{ GROUP BY DATENAME}(\text{WEEKDAY}, \text{SessionDate})$
- Average Session Length by Month = $\text{AVG}(\text{SessionLengthMin}) \text{ GROUP BY DATEPART}(\text{MONTH}, \text{SessionDate}), \text{DATEPART}(\text{YEAR}, \text{SessionDate})$
- Session Count Month-over-Month Change = $\text{COUNT}(\text{SessionID}) \text{ Current Month} - \text{COUNT}(\text{SessionID}) \text{ Previous Month}$
- Session Count Month-over-Month % Change = $((\text{Current Month} - \text{Previous Month}) / \text{Previous Month}) * 100$
- Session Count Year-over-Year Change = $\text{COUNT}(\text{SessionID}) \text{ Current Year Month} - \text{COUNT}(\text{SessionID}) \text{ Previous Year Same Month}$
- Session Count Year-over-Year % Change = $((\text{Current Year Month} - \text{Previous Year Same Month}) / \text{Previous Year Same Month}) * 100$
- Rolling 7-Day Average Sessions = $\text{AVG}(\text{COUNT}(\text{SessionID})) \text{ OVER} (\text{ORDER BY SessionDate ROWS BETWEEN 6 PRECEDING AND CURRENT ROW})$
- Rolling 30-Day Average Sessions = $\text{AVG}(\text{COUNT}(\text{SessionID})) \text{ OVER} (\text{ORDER BY SessionDate ROWS BETWEEN 29 PRECEDING AND CURRENT ROW})$
- Days Since Last Session = $\text{DATEDIFF}(\text{DAY}, \text{MAX}(\text{SessionDate}), \text{GETDATE}())$
- Active Days Count = $\text{COUNT}(\text{DISTINCT SessionDate})$
- Most Recent Session Date = $\text{MAX}(\text{SessionDate})$
- Earliest Session Date = $\text{MIN}(\text{SessionDate})$

Student Level Metrics:

- Session Count by Student Level = $\text{COUNT}(\text{SessionID}) \text{ GROUP BY StudentLevel}$
- Session Count % by Student Level = $\text{COUNT}(\text{SessionID}) * 100.0 / \text{SUM}(\text{COUNT}(\text{SessionID})) \text{ OVER}() \text{ GROUP BY StudentLevel}$

- Average Session Length by Student Level = AVG(SessionLengthMin) GROUP BY StudentLevel
- Average Satisfaction by Student Level = AVG(SatisfactionRating) GROUP BY StudentLevel
- Average Prompts by Student Level = AVG(TotalPrompts) GROUP BY StudentLevel
- AI Assistance Level Average by Student Level = AVG(AI_AssistanceLevel) GROUP BY StudentLevel
- Used Again Rate by Student Level = SUM(CASE WHEN UsedAgain = True THEN 1 ELSE 0 END) 100.0 / COUNT() GROUP BY StudentLevel
- Total Students by Student Level = COUNT(DISTINCT SessionID) GROUP BY StudentLevel
- Final Outcome Distribution by Student Level = COUNT(SessionID) GROUP BY StudentLevel, FinalOutcome

Discipline Metrics:

- Session Count by Discipline = COUNT(SessionID) GROUP BY Discipline
- Session Count % by Discipline = COUNT(SessionID) * 100.0 / SUM(COUNT(SessionID)) OVER() GROUP BY Discipline
- Average Session Length by Discipline = AVG(SessionLengthMin) GROUP BY Discipline
- Average Satisfaction by Discipline = AVG(SatisfactionRating) GROUP BY Discipline
- Average Prompts by Discipline = AVG(TotalPrompts) GROUP BY Discipline
- AI Assistance Level Average by Discipline = AVG(AI_AssistanceLevel) GROUP BY Discipline
- Used Again Rate by Discipline = SUM(CASE WHEN UsedAgain = True THEN 1 ELSE 0 END) 100.0 / COUNT() GROUP BY Discipline
- Total Students by Discipline = COUNT(DISTINCT SessionID) GROUP BY Discipline
- Final Outcome Distribution by Discipline = COUNT(SessionID) GROUP BY Discipline, FinalOutcome
- Top 3 Disciplines by Session Count = TOP 3 Discipline ORDER BY COUNT(SessionID) DESC

- Bottom 3 Disciplines by Session Count = TOP 3 Discipline ORDER BY COUNT(SessionID) ASC

Task Type Metrics:

- Session Count by Task Type = COUNT(SessionID) GROUP BY TaskType
- Session Count % by Task Type = COUNT(SessionID) * 100.0 / SUM(COUNT(SessionID)) OVER() GROUP BY TaskType
- Average Session Length by Task Type = AVG(SessionLengthMin) GROUP BY TaskType
- Average Satisfaction by Task Type = AVG(SatisfactionRating) GROUP BY TaskType
- Average Prompts by Task Type = AVG(TotalPrompts) GROUP BY TaskType
- AI Assistance Level Average by Task Type = AVG(AI_AssistanceLevel) GROUP BY TaskType
- Used Again Rate by Task Type = SUM(CASE WHEN UsedAgain = True THEN 1 ELSE 0 END) 100.0 / COUNT() GROUP BY TaskType
- Final Outcome Distribution by Task Type = COUNT(SessionID) GROUP BY TaskType, FinalOutcome
- Most Common Task Type = TOP 1 TaskType ORDER BY COUNT(SessionID) DESC
- Least Common Task Type = TOP 1 TaskType ORDER BY COUNT(SessionID) ASC

AI Assistance Level Metrics:

- Session Count by AI Assistance Level = COUNT(SessionID) GROUP BY AI_AssistanceLevel
- Session Count % by AI Assistance Level = COUNT(SessionID) * 100.0 / SUM(COUNT(SessionID)) OVER() GROUP BY AI_AssistanceLevel
- Average Session Length by AI Assistance Level = AVG(SessionLengthMin) GROUP BY AI_AssistanceLevel
- Average Satisfaction by AI Assistance Level = AVG(SatisfactionRating) GROUP BY AI_AssistanceLevel
- Average Prompts by AI Assistance Level = AVG(TotalPrompts) GROUP BY AI_AssistanceLevel

- Used Again Rate by AI Assistance Level = $\text{SUM}(\text{CASE WHEN UsedAgain} = \text{True THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$ GROUP BY AI_AssistanceLevel
- Final Outcome Distribution by AI Assistance Level = $\text{COUNT}(\text{SessionID})$ GROUP BY AI_AssistanceLevel, FinalOutcome

Final Outcome Metrics:

- Session Count by Final Outcome = $\text{COUNT}(\text{SessionID})$ GROUP BY FinalOutcome
- Session Count % by Final Outcome = $\text{COUNT}(\text{SessionID}) * 100.0 / \text{SUM}(\text{COUNT}(\text{SessionID}))$ OVER() GROUP BY FinalOutcome
- Average Session Length by Final Outcome = $\text{AVG}(\text{SessionLengthMin})$ GROUP BY FinalOutcome
- Average Satisfaction by Final Outcome = $\text{AVG}(\text{SatisfactionRating})$ GROUP BY FinalOutcome
- Average Prompts by Final Outcome = $\text{AVG}(\text{TotalPrompts})$ GROUP BY FinalOutcome
- AI Assistance Level Average by Final Outcome = $\text{AVG}(\text{AI_AssistanceLevel})$ GROUP BY FinalOutcome
- Used Again Rate by Final Outcome = $\text{SUM}(\text{CASE WHEN UsedAgain} = \text{True THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$ GROUP BY FinalOutcome
- Completion Rate = $\text{SUM}(\text{CASE WHEN FinalOutcome} = \text{'Assignment Completed'} \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$
- Abandonment Rate = $\text{SUM}(\text{CASE WHEN FinalOutcome} = \text{'Abandoned'} \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$

Used Again Metrics:

- Used Again Count (True) = $\text{SUM}(\text{CASE WHEN UsedAgain} = \text{True THEN } 1 \text{ ELSE } 0 \text{ END})$
- Used Again Count (False) = $\text{SUM}(\text{CASE WHEN UsedAgain} = \text{False THEN } 1 \text{ ELSE } 0 \text{ END})$
- Used Again Rate (% True) = $\text{SUM}(\text{CASE WHEN UsedAgain} = \text{True THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$
- Average Satisfaction when Used Again = True = $\text{AVG}(\text{SatisfactionRating})$ WHERE UsedAgain = True

- Average Satisfaction when Used Again = False = $\text{AVG}(\text{SatisfactionRating})$ WHERE UsedAgain = False
- Average Session Length when Used Again = True = $\text{AVG}(\text{SessionLengthMin})$ WHERE UsedAgain = True
- Average Session Length when Used Again = False = $\text{AVG}(\text{SessionLengthMin})$ WHERE UsedAgain = False
- Average Prompts when Used Again = True = $\text{AVG}(\text{TotalPrompts})$ WHERE UsedAgain = True
- Average Prompts when Used Again = False = $\text{AVG}(\text{TotalPrompts})$ WHERE UsedAgain = False
- AI Assistance Level Average when Used Again = True = $\text{AVG}(\text{AI_AssistanceLevel})$ WHERE UsedAgain = True
- AI Assistance Level Average when Used Again = False = $\text{AVG}(\text{AI_AssistanceLevel})$ WHERE UsedAgain = False

Satisfaction Rating Metrics:

- Satisfaction Distribution = $\text{COUNT}(\text{SessionID})$ GROUP BY SatisfactionRating
- Satisfaction % per Rating = $\text{COUNT}(\text{SessionID}) * 100.0 / \text{SUM}(\text{COUNT}(\text{SessionID}))$ OVER() GROUP BY SatisfactionRating
- High Satisfaction Rate = $\text{SUM}(\text{CASE WHEN SatisfactionRating} \geq 4 \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$ 100.0 / COUNT()
- Low Satisfaction Rate = $\text{SUM}(\text{CASE WHEN SatisfactionRating} \leq 2 \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$ 100.0 / COUNT()
- Neutral Satisfaction Rate = $\text{SUM}(\text{CASE WHEN SatisfactionRating} = 3 \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) / \text{COUNT}()$ 100.0 / COUNT()
- Average Satisfaction by Discipline = $\text{AVG}(\text{SatisfactionRating})$ GROUP BY Discipline
- Average Satisfaction by Task Type = $\text{AVG}(\text{SatisfactionRating})$ GROUP BY TaskType
- Average Satisfaction by Student Level = $\text{AVG}(\text{SatisfactionRating})$ GROUP BY StudentLevel
- Average Satisfaction by AI Assistance Level = $\text{AVG}(\text{SatisfactionRating})$ GROUP BY AI_AssistanceLevel

- Average Satisfaction by Final Outcome = AVG(SatisfactionRating) GROUP BY FinalOutcome
- Satisfaction Trend by Month = AVG(SatisfactionRating) GROUP BY DATEPART(MONTH, SessionDate), DATEPART(YEAR, SessionDate)

Engagement Metrics:

- Average Session Length per Student = AVG(SessionLengthMin) GROUP BY SessionID
- Average Prompts per Student = AVG(TotalPrompts) GROUP BY SessionID
- Total Sessions per Student = COUNT(SessionID) GROUP BY SessionID
- Students with Single Session = COUNT(DISTINCT SessionID) WHERE SessionCount = 1
- Students with Multiple Sessions = COUNT(DISTINCT SessionID) WHERE SessionCount > 1
- Returning Student Rate = COUNT(DISTINCT SessionID WHERE SessionCount > 1) * 100.0 / COUNT(DISTINCT SessionID)
- Sessions per Returning Student Average = AVG(SessionCount) WHERE SessionCount > 1
- Average Days Between Sessions per Student = DATEDIFF(DAY, MIN(SessionDate), MAX(SessionDate)) / (COUNT(SessionID) - 1) GROUP BY SessionID
- Student Retention Rate = COUNT(DISTINCT SessionID WHERE SessionCount > 1) * 100.0 / COUNT(DISTINCT SessionID)

Performance/Efficiency Metrics:

- Prompts per Minute = TotalPrompts / SessionLengthMin
- Average Prompts per Minute = AVG(TotalPrompts / SessionLengthMin)
- Sessions with Prompts per Minute > Threshold = COUNT(SessionID) WHERE (TotalPrompts / SessionLengthMin) > Threshold
- Average Session Length per Prompts = AVG(SessionLengthMin / TotalPrompts)
- Satisfaction per Minute of Session = SatisfactionRating / SessionLengthMin

Combined/Cross Metrics:

- Satisfaction by Student Level and Discipline = AVG(SatisfactionRating) GROUP BY StudentLevel, Discipline
- Satisfaction by Task Type and AI Assistance Level = AVG(SatisfactionRating) GROUP BY TaskType, AI_AssistanceLevel
- Used Again Rate by Student Level and Discipline = SUM(CASE WHEN UsedAgain = True THEN 1 ELSE 0 END) 100.0 / COUNT() GROUP BY StudentLevel, Discipline
- Used Again Rate by Task Type and AI Assistance Level = SUM(CASE WHEN UsedAgain = True THEN 1 ELSE 0 END) 100.0 / COUNT() GROUP BY TaskType, AI_AssistanceLevel
- Average Session Length by Discipline and Task Type = AVG(SessionLengthMin) GROUP BY Discipline, TaskType
- Average Prompts by Student Level and Task Type = AVG(TotalPrompts) GROUP BY StudentLevel, TaskType
- Session Count by Discipline and Task Type = COUNT(SessionID) GROUP BY Discipline, TaskType
- Session Count by Student Level and Final Outcome = COUNT(SessionID) GROUP BY StudentLevel, FinalOutcome
- Satisfaction by AI Assistance Level and Final Outcome = AVG(SatisfactionRating) GROUP BY AI_AssistanceLevel, FinalOutcome
- Used Again Rate by Final Outcome and Satisfaction Level = SUM(CASE WHEN UsedAgain = True THEN 1 ELSE 0 END) 100.0 / COUNT() GROUP BY FinalOutcome, CASE WHEN SatisfactionRating >= 4 THEN 'High' WHEN SatisfactionRating <= 2 THEN 'Low' ELSE 'Neutral' END

Cumulative/Running Metrics:

- Cumulative Sessions (YTD) = SUM(COUNT(SessionID)) OVER (ORDER BY SessionDate)
- Cumulative Total Session Length (YTD) = SUM(SessionLengthMin) OVER (ORDER BY SessionDate)
- Cumulative Total Prompts (YTD) = SUM(TotalPrompts) OVER (ORDER BY SessionDate)

- Running Average Satisfaction = $\text{AVG}(\text{SatisfactionRating}) \text{ OVER } (\text{ORDER BY SessionDate})$
- Running Average Session Length = $\text{AVG}(\text{SessionLengthMin}) \text{ OVER } (\text{ORDER BY SessionDate})$

Ranking/Top N Metrics:

- Top 10% Students by Session Count = $\text{PERCENTILE}(\text{COUNT}(\text{SessionID}) \text{ GROUP BY SessionID}, 0.9)$
- Top 10% Students by Total Session Length = $\text{PERCENTILE}(\text{SUM}(\text{SessionLengthMin}) \text{ GROUP BY SessionID}, 0.9)$
- Top 10% Students by Average Satisfaction = $\text{PERCENTILE}(\text{AVG}(\text{SatisfactionRating}) \text{ GROUP BY SessionID}, 0.9)$
- Bottom 10% Students by Average Satisfaction = $\text{PERCENTILE}(\text{AVG}(\text{SatisfactionRating}) \text{ GROUP BY SessionID}, 0.1)$
- Top Disciplines by Average Satisfaction = TOP N Discipline ORDER BY $\text{AVG}(\text{SatisfactionRating}) \text{ DESC}$
- Bottom Disciplines by Average Satisfaction = TOP N Discipline ORDER BY $\text{AVG}(\text{SatisfactionRating}) \text{ ASC}$
- Top Task Types by Session Count = TOP N TaskType ORDER BY $\text{COUNT}(\text{SessionID}) \text{ DESC}$
- Top AI Assistance Levels by Satisfaction = TOP N AI_AssistanceLevel ORDER BY $\text{AVG}(\text{SatisfactionRating}) \text{ DESC}$

Derived KPIs:

- Satisfaction Score Index = $(\text{AVG}(\text{SatisfactionRating}) / 5.0) * 100$
- Engagement Score = $(\text{COUNT}(\text{SessionID}) / \text{MAX}(\text{COUNT}(\text{SessionID}))) 0.33 + (\text{SUM}(\text{SessionLengthMin}) / \text{MAX}(\text{SUM}(\text{SessionLengthMin}))) 0.33 + (\text{SUM}(\text{TotalPrompts}) / \text{MAX}(\text{SUM}(\text{TotalPrompts}))) * 0.34$
- AI Adoption Rate = $\text{SUM}(\text{CASE WHEN AI_AssistanceLevel} \geq 3 \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) 100.0 / \text{COUNT}()$
- High Performer Rate = $\text{SUM}(\text{CASE WHEN SatisfactionRating} \geq 4 \text{ AND FinalOutcome} = \text{'Assignment Completed'} \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) 100.0 / \text{COUNT}()$
- At-Risk Student Rate = $\text{SUM}(\text{CASE WHEN SatisfactionRating} \leq 2 \text{ AND UsedAgain} = \text{False} \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) 100.0 / \text{COUNT}()$

- Session Productivity Score = $(\text{AVG}(\text{TotalPrompts} / \text{SessionLengthMin}) * \text{AVG}(\text{SatisfactionRating}))$
- Student Loyalty Index = $(\text{SUM}(\text{CASE WHEN UsedAgain} = \text{True THEN } 1 \text{ ELSE } 0 \text{ END}) * 0.4 + \text{SUM}(\text{CASE WHEN SatisfactionRating} \geq 4 \text{ THEN } 1 \text{ ELSE } 0 \text{ END}) * 0.3 + \text{SUM}(\text{CASE WHEN COUNT(SessionID) > 1 THEN } 1 \text{ ELSE } 0 \text{ END}) * 0.3) / \text{COUNT}(\text{DISTINCT SessionID}) * 100$